

2.2 Group 2

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Easy

Time allowed: 60

Score: /49

Percentage: /100

Question 1a

This question is about the reactions of Group 2 elements.

i)

Give the names of a Group 2 element that burns in air to give a white flame.

[1]

ii)

Write the balanced chemical equation for the reaction of this Group 2 element in air.

[2]

[3 marks]

Question 1b

i)

Give the reagent and condition required for magnesium to form magnesium hydroxide.

[2]

ii)

Write the equation for this reaction. Your answer should include state symbols.

[1]

[3 marks]

Question 1c

Calcium is added to hydrochloric acid.

i)

Give **one** expected observation for this reaction.

[1]

ii)

Write the balanced chemical equation for the reaction of calcium with hydrochloric acid. Your answer should include state symbols.

[2]

[3 marks]

Question 1d

Compare the reactions of barium with hydrochloric acid and sulfuric acid. Your answer should include chemical equations with state symbols for each reaction.

[5 marks]

Question 2a

Group 2 oxides can react with water to form Group 2 hydroxides.

Describe the general trends in solubility and pH of the Group 2 hydroxides as you move down Group 2.

[2 marks]

Question 2b

3.71 g of calcium hydroxide, Ca(OH)_2 , is added to hydrochloric acid.

i)

Write the balanced chemical equation for this reaction.

[2]

ii)

Calculate the mass of calcium chloride produced in this reaction.

[2]

[4 marks]

Question 2c

Strontium carbonate reacts with hydrochloric acid to form strontium chloride, water and carbon dioxide.

i)

Describe how you could prove that the gas produced is carbon dioxide.

[2]

ii)

Carbon dioxide is also produced when strontium carbonate reacts with sulfuric acid. The effervescence observed in this reaction stops after a short period of time.

Suggest why the reaction stops even though the reactants have not been used up.

[2]

[4 marks]

Question 3a

State the type of reaction that occurs when strontium carbonate is heated to form strontium oxide and carbon dioxide.

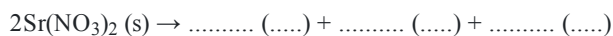


[1 mark]

Question 3b

When heated, strontium nitrate breaks down to form strontium oxide, nitrogen dioxide and oxygen.

Complete the balanced chemical equation for this reaction.

**[2 marks]****Question 3c**

5.00 g of magnesium carbonate breaks down to form 1.75 g magnesium oxide and carbon dioxide when heated.



Calculate the percentage yield of magnesium oxide. Show your working.

[3 marks]**Question 3d**

Describe the trend in the thermal stability of the Group 2 carbonates and nitrates as you move down Group 2.

[1 mark]

Question 4a

The Group 2 elements can be used to show the trends in properties down a group in the Periodic Table.

Complete Table 4.1 to describe the general trends as you move down Group 2.

Table 4.1

Property	Trend
Melting point	
Atomic radius	
First ionisation energy	

[3 marks]**Question 4b**

Name the Group 2 element that does not follow the general trend in melting point.

[1 mark]**Question 4c**

Write chemical equations for the first ionisation energy of barium and the second ionisation energy of calcium.

[2 marks]

Question 4d

The trend in ionisation energy down Group 2 can be explained by the factors shown in Table 4.2.

Complete the table stating how each factor changes down Group 2.

Table 4.2

Factor affecting ionisation energy	Increase / decrease
Distance between the nucleus and outer electron	
Shielding	
Nuclear attraction	

[3 marks]

Question 5a

A group of students are asked to prepare a sample of barium sulfate using barium carbonate.

Student **A** suggests adding sulfuric acid to the barium carbonate.

i)

Write the balanced chemical equation for the reaction of barium carbonate and sulfuric acid.

[1]

ii)

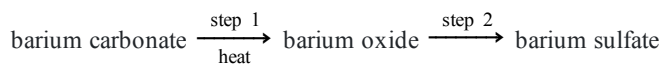
Suggest why this preparation of barium sulfate will not have a 100% yield. Explain your answer.

[2]

[3 marks]

Question 5b

Student **B** suggests that barium carbonate can be converted into barium sulfate in a two-step process.



i)

Name the type of reaction that occurs in step 1.

[1]

ii)

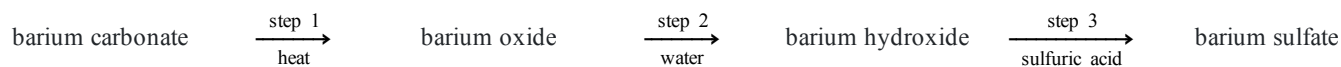
In step 2, barium oxide is reacted with sulfuric acid. Write the chemical equation for the reaction in step 2.

[1]

[2 marks]

Question 5c

Student **C** makes an adaptation to student **B**'s suggestion.



Describe the solubility in water of barium hydroxide and barium sulfate.

[2 marks]

Question 5d

Write the ionic equation for the reaction in step 3 between barium hydroxide and sulfuric acid to form barium sulfate. Include state symbols.

[2 marks]

Model Answers: Easy

1a

a)

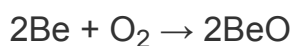
i) A Group 2 element that burns in air to give a white flame is:

- Beryllium / Be

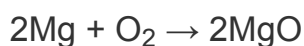
OR

Magnesium / Mg; [1 mark]

ii) The balanced chemical equation for the reaction of the Group 2 element in air is:



OR



- Correct chemical formulae; [1 mark]
- Correct balancing; [1 mark]

[Total: 3 marks]

- Beryllium and magnesium both burn in air with a white flame
 - Magnesium is well known
 - Beryllium is less well known
 - It can be ignited but it is difficult to actually burn
- The reaction in air is combustion
 - The equation given must match the element stated as an answer in part (i)
 - Magnesium + oxygen → magnesium oxide
 - $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$
 - There are two oxygen atoms on the left-hand side so you need 2MgO
 - $\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
 - There are now two magnesium atoms on the right-hand side so you need 2Mg
 - $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
 - The same process can be applied to balancing the equation for the combustion of

beryllium

- $\text{Be} + \text{O}_2 \rightarrow \text{BeO}$
- $\text{Be} + \text{O}_2 \rightarrow 2\text{BeO}$
- $2\text{Be} + \text{O}_2 \rightarrow 2\text{BeO}$

1b

b)

i) The reagent and condition required for magnesium to react with water are:

- Reagent = water; [1 mark]
- Condition = the water must be gaseous / steam; [1 mark]

ii) The equation for the reaction is:

- $\text{Mg (s)} + 2\text{H}_2\text{O (g)} \rightarrow \text{Mg(OH)}_2 \text{ (s)} + \text{H}_2 \text{ (g)}$; [1 mark]

[Total: 3 marks]

- The reaction of magnesium with water is very slow
- When thermal energy is added by making the water gaseous, the reaction proceeds quickly and violently
- Like most Group 2 metals, the reaction forms the metal hydroxide and hydrogen according to the equation
 - $\text{M} + 2\text{H}_2\text{O} \rightarrow \text{M(OH)}_2 + \text{H}_2$
- **Careful:** The water must be gaseous / (g) for this reaction **AND** the magnesium hydroxide formed is insoluble / (s)
 - You must give state symbols to achieve the mark for the equation

1c

c)

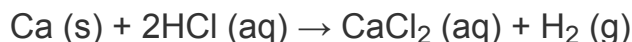
i) **One** expected observation for the reaction of calcium with hydrochloric acid is:

- Fizzing / bubbling / effervescence

OR

Smoke forms; [1 mark]

ii) The balanced chemical equation for the reaction of calcium with hydrochloric acid is:



- Correct chemical formulae; [1 mark]
- Correct balancing

AND

All state symbols correct; [1 mark]

[Total: 3 marks]

- The reaction of calcium with hydrochloric acid is similar to the reaction of magnesium with hydrochloric acid
 - Apart from the obvious change in reactant and product, the only noticeable difference is how vigorous the reaction is
- **Remember:** Metal + acid $\rightarrow$ salt + hydrogen
 - Calcium + hydrochloric acid $\rightarrow$ calcium chloride + hydrogen
- The unbalanced symbol equation is:
 - $\text{Ca} + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$
- There are two chlorine atoms and two hydrogen atoms on the right-hand side, which means that you need 2HCl
 - $\text{Ca} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2$
- You should know that calcium is solid / (s), hydrochloric acid is aqueous / (aq) and hydrogen is gaseous / (g)
- Calcium chloride is soluble, which means that it is aqueous / (aq)
 - $\text{Ca (s)} + 2\text{HCl (aq)} \rightarrow \text{CaCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$

1d

d) To compare the reactions of barium with hydrochloric acid and sulfuric acid:

Barium and hydrochloric acid equation:

- $\text{Ba (s)} + 2\text{HCl (aq)} \rightarrow \text{BaCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$; [1 mark]

Barium and sulfuric acid equation:

- $\text{Ba (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{BaSO}_4 \text{ (s)} + \text{H}_2 \text{ (g)}$; [1 mark]

Comparison:

- Both reactions form hydrogen gas / $\text{H}_2(\text{g})$; [1 mark]
 - The reaction with hydrochloric acid / HCl requires 2 moles of hydrochloric acid / HCl (compared to one mole of sulfuric acid / H_2SO_4); [1 mark]
 - The reaction with hydrochloric acid forms an aqueous solution
- AND**
- The reaction with sulfuric acid forms a solid / precipitate; [1 mark]

[Total: 5 marks]

- **Remember:** Hydrochloric acid forms chloride salts and sulfuric acid forms sulfate salts
- Both reactions follow the same general equation:
 - $\text{Metal} + \text{acid} \rightarrow \text{salt} + \text{hydrogen}$
 - The key similarity is the production of hydrogen gas
- **Tip:** When you are asked to compare and you are talking about the differences of both things that you are comparing
 - Don't just talk about one and assume the examiner knows the other half, the examiner wants to see that you know it too
 - The differences are with the amount of acid used and the physical state of the product

2a

a) The general trends in solubility and pH of the Group 2 hydroxides as you move down Group 2 are:

- The solubility increases
- OR**
- The (Group 2) hydroxides become more soluble; [1 mark]
- The pH increases
- OR**
- The solutions become more alkaline; [1 mark]

[Total: 2 marks]

- You should know that magnesium hydroxide is insoluble / sparingly soluble in water and barium hydroxide is soluble in water
- Therefore, the solubility increases as you move down Group 2
- This means that there are more hydroxide ions / OH^- in the solution as you move down

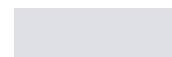
the group

- As a result of this, the pH increases
 - **Careful:** Students often get mixed up when talking about pH
 - When the pH decreases, the acidity increases i.e. it becomes more acidic / less alkaline
 - When the pH increases, the acidity decreases i.e. it becomes more alkaline / less acidic

2b

b)

i) The balanced chemical equation for the reaction of calcium hydroxide with hydrochloric acid is:



The (Group 2) hydroxides become more soluble; [1 mark]

- The pH increases

OR

The solutions become more alkaline; [1 mark]

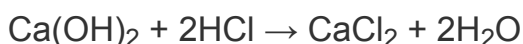
[Total: 2 marks]

- You should know that magnesium hydroxide is insoluble / sparingly soluble in water and barium hydroxide is soluble in water
- Therefore, the solubility increases as you move down Group 2
- This means that there are more hydroxide ions / OH^- in the solution as you move down the group
- As a result of this, the pH increases
 - **Careful:** Students often get mixed up when talking about pH
 - When the pH decreases, the acidity increases i.e. it becomes more acidic / less alkaline
 - When the pH increases, the acidity decreases i.e. it becomes more alkaline / less acidic

2b

b)

i) The balanced chemical equation for the reaction of calcium hydroxide with hydrochloric acid is:



- Correct chemical formulae; [1 mark]
- Correct balancing; [1 mark]

ii) To calculate the mass of calcium chloride produced in this reaction:

- Formula mass of $\text{Ca(OH)}_2 = 74.1$

AND

Moles of $\text{Ca(OH)}_2 = 0.05$; [1 mark]

- Formula mass of $\text{CaCl}_2 = 111.1$

AND

Mass of CaCl_2 produced = 5.56 (g); [1 mark]

[Total: 4 marks]

- **Remember:** Metal hydroxide + acid → metal salt + water
 - Hydrochloric acid makes chloride salts
 - Calcium hydroxide + hydrochloric acid → calcium chloride + water
 - $\text{Ca(OH)}_2 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$
 - There are two chlorine atoms on the right-hand side, so you need 2HCl
 - $\text{Ca(OH)}_2 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$
 - This has also balanced the entire equation
- To calculate the mass of calcium chloride produced, you need to:
 - Calculate the relative formula masses of calcium hydroxide and calcium chloride
 - $\text{Ca(OH)}_2 = 40.1 + ((16.0 + 1.0) \times 2) = 74.1$
 - $\text{CaCl}_2 = 40.1 + (35.5 \times 2) = 111.1$
 - The moles of $\text{Ca(OH)}_2 = \frac{\text{mass}}{M_r} = \frac{3.71}{74.1} = 0.05$
 - One mole of Ca(OH)_2 forms one mole of CaCl_2
 - Therefore, there are also 0.05 moles of CaCl_2
 - Mass of $\text{CaCl}_2 = \text{moles} \times M_r = 0.05 \times 111.1 = 5.56 \text{ g}$

2c

c)

i) You could prove that the gas produced is carbon dioxide by:

- Bubbling / passing the gas through limewater; [1 mark]
- Which turns cloudy / milky

OR

Which forms a white precipitate; [1 mark]

ii) The reaction stops even though the reactants have not been used up because:

- Insoluble strontium sulfate is formed; [1 mark]
- Which coats the strontium carbonate (stopping the reaction); [1 mark]

[Total: 4 marks]

- You should know the test for carbon dioxide from your previous studies
 - Any A-level course can ask questions from your prior learning in Chemistry

- The reactions of calcium carbonate, strontium carbonate and barium carbonate with sulfuric acid all form precipitates of their respective sulfates
 - These sulfates can then coat the carbonate
 - This means that the sulfate is acting as a physical barrier between the carbonate and the acid, resulting in the reaction appearing to stop even though it is not complete

3a

a) The type of reaction that occurs when strontium carbonate is heated to form strontium oxide and carbon dioxide is:

- Thermal decomposition; [1 mark]

[Total: 1 mark]

- The question tells you that the reaction is heated
 - Therefore, it is a 'thermal' reaction
- The question and equation show that the calcium carbonate is being broken down
 - Therefore, it is a 'decomposition' reaction
- Both thermal **AND** decomposition are required for the mark

3b

b) The completed chemical equation is:



- Correct chemical formulae
AND
Correct state symbols; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- The strontium ion has a 2+ and the oxide ion has a 2- charge
 - Therefore, the formula for strontium oxide is SrO
 - Strontium oxide is a solid
- You should know that nitrogen dioxide is NO₂ and oxygen is O₂

- Nitrogen dioxide and oxygen are both gases
- Adding this information to the equation gives the following unbalanced equation
 - $2\text{Sr}(\text{NO}_3)_2 (\text{s}) \rightarrow \text{SrO} (\text{s}) + \text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$
- There are two strontium atoms on the left-hand side, so we need to produce 2SrO
 - $2\text{Sr}(\text{NO}_3)_2 (\text{s}) \rightarrow 2\text{SrO} (\text{s}) + \text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$
- There are four nitrogen atoms on the left-hand side, so we need to produce 4NO₂
 - $2\text{Sr}(\text{NO}_3)_2 (\text{s}) \rightarrow 2\text{SrO} (\text{s}) + 4\text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$
- The equation is now balanced

3c

c) To calculate the percentage yield of magnesium oxide:

- Formula mass of $\text{MgCO}_3 = 84.3$
AND
Moles of $\text{MgCO}_3 = 0.059$; [1 mark]
- Formula mass of $\text{MgO} = 40.3$
AND
Theoretical mass of $\text{MgO} = 2.39 \text{ (g)}$; [1 mark]
- Percentage yield = 73.2%; [1 mark]

[Total: 3 marks]

- **Remember:** Percentage yield = $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100$
- The actual yield of magnesium oxide is 1.75 g
- To calculate the theoretical yield of magnesium, you have to:
 - Calculate the relative formula mass of magnesium carbonate
 - $\text{MgCO}_3 = 24.3 + 12.0 + (16.0 \times 3) = 84.3$
 - Calculate the moles of magnesium carbonate
 - Moles = $\frac{\text{mass}}{M_r} = \frac{5.00}{84.3} = 0.059$
 - One mole of magnesium carbonate forms one mole of magnesium oxide
 - This means that 0.059 moles of magnesium oxide are formed
 - Calculate the formula mass of magnesium oxide
 - $\text{MgO} = 24.3 + 16.0 = 40.3$
 - Calculate the theoretical mass of magnesium oxide
 - Mass = moles $\times M_r = 0.059 \times 40.3 = 2.39 \text{ g}$

- Calculate the percentage yield

- $\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{1.75}{2.39} \times 100 = 73.2\%$

3d

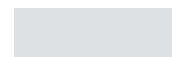
d) The trend in the thermal stability of the Group 2 carbonates and nitrates as you move down Group 2 is:

- The thermal stability increases (as you move down Group 2); [1 mark]

[Total: 1 mark]

- You should be able to describe and, also, explain this trend

4a



- Calculate the relative formula mass of magnesium carbonate
 - $\text{MgCO}_3 = 24.3 + 12.0 + (16.0 \times 3) = 84.3$
- Calculate the moles of magnesium carbonate
 - $\text{Moles} = \frac{\text{mass}}{M_r} = \frac{5.00}{84.3} = 0.059$
- One mole of magnesium carbonate forms one mole of magnesium oxide
 - This means that 0.059 moles of magnesium oxide are formed
- Calculate the formula mass of magnesium oxide
 - $\text{MgO} = 24.3 + 16.0 = 40.3$
- Calculate the theoretical mass of magnesium oxide
 - $\text{Mass} = \text{moles} \times M_r = 0.059 \times 40.3 = 2.39 \text{ g}$
- Calculate the percentage yield
 - $\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{1.75}{2.39} \times 100 = 73.2\%$

3d

d) The trend in the thermal stability of the Group 2 carbonates and nitrates as you move down Group 2 is:

- The thermal stability increases (as you move down Group 2); [1 mark]

[Total: 1 mark]

- You should be able to describe and, also, explain this trend

4a

a) The general trends as you move down Group 2 are:

- Melting point = decreases; [1 mark]
- First ionisation energy = decreases; [1 mark]
- Atomic radius = increases; [1 mark]

[Total: 3 marks]

- You need to be able to describe and, also, explain the trends as you move down Group 2

4b

b) The Group 2 element that does not follow the general trend in melting point is

- Magnesium / Mg; [1 mark]

[Total: 1 mark]

- You are not expected to know the reason for the deviation of magnesium

4c

c) The chemical equations for the first ionisation energy of barium and the second ionisation energy of calcium are:

- $\text{Ba (g)} \rightarrow \text{Ba}^+ \text{ (g)} + \text{e}^-$; [1 mark]
- $\text{Ca}^+ \text{ (g)} \rightarrow \text{Ca}^{2+} \text{ (g)} + \text{e}^-$; [1 mark]

[Total: 2 marks]

- **Remember:** First ionisation energy is the energy required to remove one mole of electrons from one mole of gaseous atoms
 - $\text{X (g)} \rightarrow \text{X}^+ \text{ (g)} + \text{e}^-$;
- **Remember:** Second ionisation energy is the energy required to remove one mole of electrons from one mole of gaseous 1+ ions
 - $\text{X}^+ \text{ (g)} \rightarrow \text{X}^{2+} \text{ (g)} + \text{e}^-$

4d

d) The points to explain the trend in first ionisation energies down Group 2 are:

- Distance between the nucleus and outer electron = increase; [1 mark]
- Shielding = increase; [1 mark]
- Nuclear attraction = decrease; [1 mark]

[Total: 3 marks]

- The distance between the nucleus and outer electron and the amount of shielding are two points that you should comment on for this type of question

- You can then link these two points to the overall force of nuclear attraction to give your final answer
- When asked to explain the trend in ionisation energy, you should always refer to these three factors
 - State the trend and then talk about the distance between the nucleus and the outer e^- and the amount of shielding first, before linking these to nuclear attraction
 - For example:
 - The first ionisation energy decreases down Group 2
 - This is because the distance between the nucleus and outer electrons increases as well as the amount of shielding increasing which means that the overall force of nuclear attraction decreases, i.e. less energy is required to remove the outer electron

5a

a)

i) The balanced chemical equation for the reaction of barium carbonate and sulfuric acid is:



ii) This preparation of barium sulfate will not have a 100% yield because:

- Not all the barium carbonate reacts; [1 mark]
- (Because,) a layer of barium sulfate coats the barium carbonate / stops the barium carbonate reacting; [1 mark]

[Total: 3 marks]

- **Remember:** Metal carbonate + acid $\rightarrow$ Metal salt + water + carbon dioxide
 - Sulfuric acid forms sulfates
- Barium sulfate is the desired product
 - However, barium sulfate is insoluble and coats any remaining barium carbonate
 - This stops the barium carbonate from being able to fully react with the sulfuric acid

5b

b)

i) The type of reaction that occurs in step 1 is

- Thermal decomposition; [1 mark]

ii) The chemical equation for the reaction in step 2 is:

- $\text{BaO} + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}$; [1 mark]

[Total: 2 marks]

- In step 1, barium carbonate is heated and breaks down to form barium oxide and carbon dioxide
 - $\text{BaCO}_3 \rightarrow \text{BaO} + \text{CO}_2$
 - Using heat to break a compound down is thermal decomposition
- **Remember:** Metal oxide + sulfuric acid $\rightarrow$ metal sulfate + water

5c

c) The correct solubilities in water are:

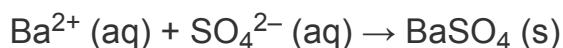
- Barium hydroxide = soluble; [1 mark]
- Barium sulfate = insoluble; [1 mark]

[Total: 2 marks]

- You should know the trend in solubility of the Group 2 hydroxides and sulfates
 - The hydroxides become more soluble as you move down the group
 - The sulfates become less soluble as you move down the group
- Since barium is near the bottom of Group 2
 - Barium hydroxide will be soluble
 - Barium sulfate will be insoluble

5d

d) The ionic equation for the reaction in step 3 between barium hydroxide and sulfuric acid to form barium sulfate is:



- Correct chemical formulae; [1 mark]

- Correct state symbols; [1 mark]

[Total: 2 marks]

- The reaction taking place in step 3 is:
 - $\text{Ba}(\text{OH})_2 (\text{aq}) + \text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{BaSO}_4 (\text{s}) + \text{H}_2\text{O} (\text{l})$
- **Remember:** Spectator ions are not included in ionic equations
 - This means that the following ions and chemicals should be removed
 - The hydroxide ions from barium hydroxide
 - The hydrogen ions from sulfuric acid
 - The water
- This leaves the correct ionic equation
 - $\text{Ba}^{2+} (\text{aq}) + \text{SO}_4^{2-} (\text{aq}) \rightarrow \text{BaSO}_4 (\text{s})$

